

JURONG JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATIONS
Higher 2

CANDIDATE
NAME

CLASS

BIOLOGY

9648/03

Applications Paper and Planning Question

13 September 2016

Paper 3

2 hours

Additional Materials: Answer Booklet/Paper

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.

Answer **all** questions.

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
1	
2	
3	
4	
5	
Total	

This document consists of **15** printed pages and **1** blank page.

[Turn over]

Answer **all** the questions in this section.

- 1 Fig. 1.1 shows the plasmid pOLUC. Plasmid pOLUC is a cloning vector used in creating recombinant DNA. It contains two selectable marker genes, luciferase gene and ampicillin resistance gene, Amp^R. A multiple cloning site (MCS) is located within the luciferase gene. Luciferase gene, when expressed, hydrolyses luciferin which results in a glowing bacterial colony.

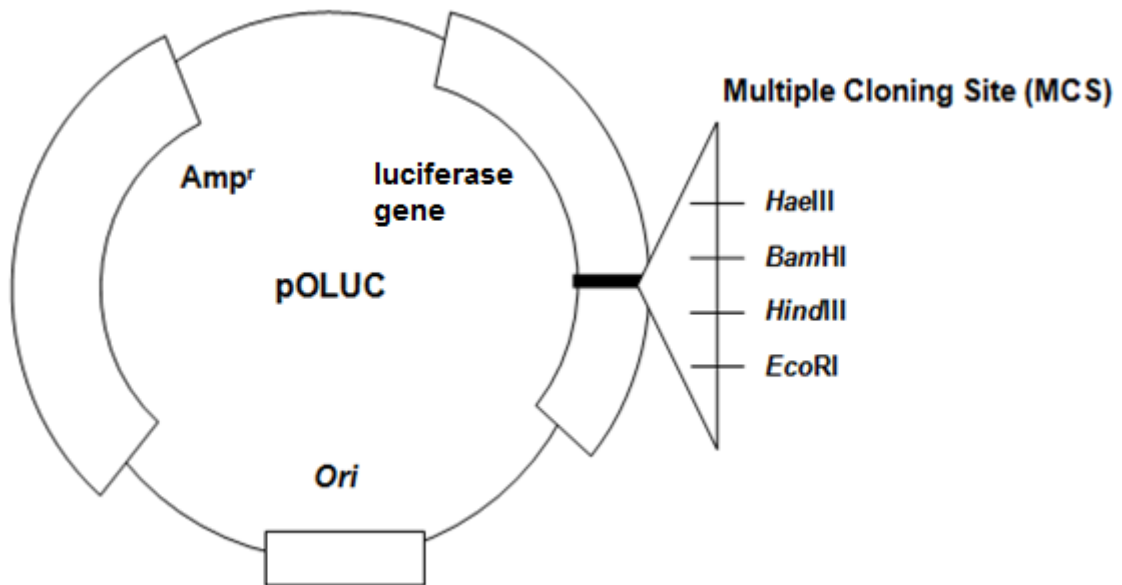


Fig. 1.1

- (a) Describe and explain one property of pOLUC that allows it to be used as DNA cloning vector. [2]

Fig. 1.2 shows the target DNA sequence that will be inserted into the MCS of pOLUC.

5' – GGATCCATCCCGATCGATCCTTTAAGCTAGCTAGGATCC – 3'

3' – CCTAGGTAGGGCTAGCTAGGAAATTCGATCGATCCTAGG – 5'

Fig. 1.2

Fig. 1.3 shows the types of restriction enzymes that can be used to cleave pOLUC at the Multiple Cloning Site (MCS) shown in Fig. 1.1.

Restriction enzyme Specific target base sequence of DNA

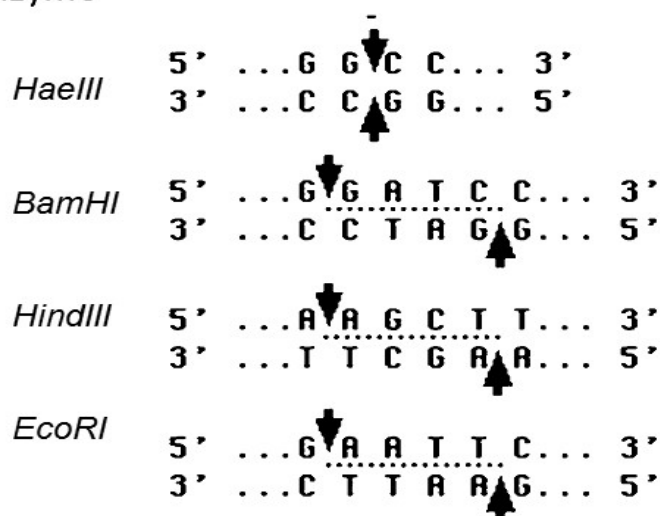


Fig. 1.3

(b) Describe the natural function of a restriction enzyme. [2]

(c) State the restriction enzyme used to cleave the target DNA sequence before it is inserted into vector pOLUC. Give a reason for your choice. [2]

(d) Outline the steps in producing a recombinant plasmid. [3]

(e) Explain how bacteria with recombinant plasmids are selected from bacteria that are transformed. [5]

(f) Suggest why replica plating was not necessary in this experiment. [1]

[Total: 15]

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- 2 Cystic fibrosis (CF) is a genetic disorder caused by mutations in the gene that code for the CF transmembrane conductance regulator (CFTR) that contributes to regulation of ion flux at the membrane of epithelial cells.

A DNA sequence, KM-19 found within 1 centiMorgan of the CFTR gene has been identified. This segment exhibits restriction fragment length polymorphism (RFLP) that can be used as a RFLP marker for the genetic screening of CF.

DNA samples containing the CFTR gene and KM-19 from four individuals were treated with restriction enzyme, *EcoRI* and subjected to gel electrophoresis.

(a) Outline why gel electrophoresis separates DNA fragments. [3]

Fig. 2.1 shows the positions of various restriction fragment of KM-19 and a probe that was used to detect the polymorphism. RFLP variant 1 of KM-19 was associated with the normal CFTR allele while RFLP variant 2 of KM-19 was linked to the mutant CFTR allele.

It was observed that individual C does not show the expected band pattern for affected individual with CF.

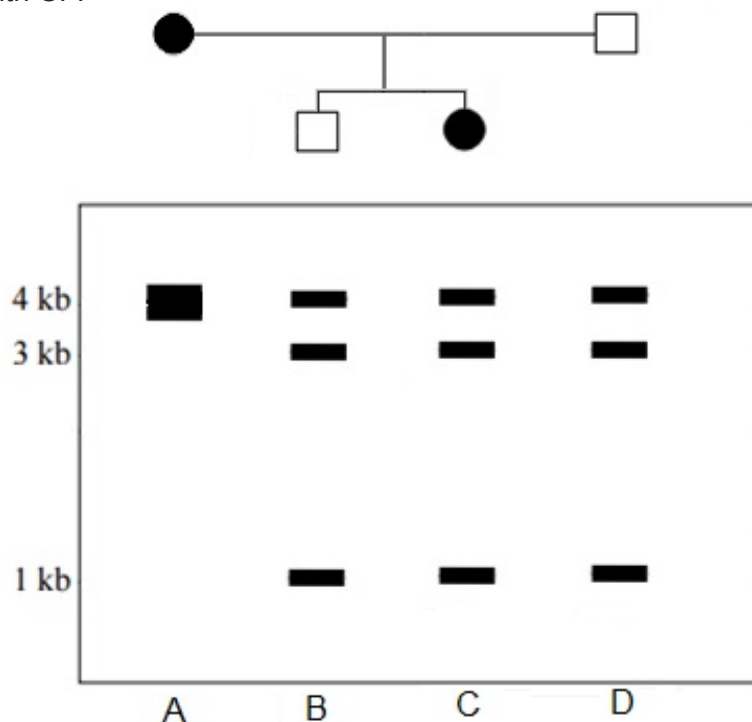


Fig. 2.1

(b) With reference to Fig. 2.1,

(i) indicate the restriction site(s) of *EcoRI* and DNA fragment sizes on RFLP variants 1 and 2 of KM-19; [1]

(ii) draw on the restriction map where the probe binds. [1]



(c) Explain why genetic markers like KM-19 can be used to detect the presence of disease-causing alleles. [2]

(d) Suggest an explanation for the discrepancy observed in C regarding the band pattern and occurrence of CF. [3]

(e) In the absence of gel electrophoresis and nucleic acid hybridisation, outline a possible method in detecting RFLP variant 2 of KM-19. [1]

[Total: 11]

- 3 Severe combined immunodeficiency (SCID) is a genetic disorder whereby the affected individuals have non-functional immune system with increased vulnerability to infections. The most common form of SCID is X-linked SCID, caused by mutation of interleukin-2 receptor gamma (IL2RG) gene on X chromosome.

Fig. 3.1 shows interleukin-2 receptor that is involved in JAK-STAT signalling pathway in the T cells of a healthy individual.

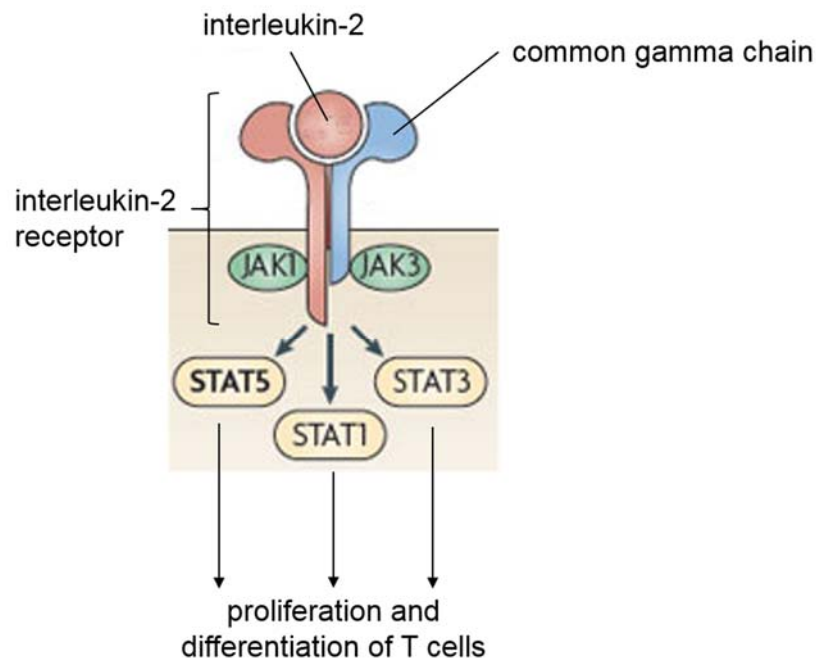


Fig. 3.1

- (a) With reference to Fig. 3.1 and the information above, describe the mutation that results in X-linked SCID. [3]

In 2002, two young individuals with X-linked SCID were treated with retroviral-based gene therapy on their isolated blood stem cells.

(b) Outline how a retroviral vector delivers normal functional IL2RG allele into the blood stem cells. [3]

Fig. 3.2 shows the average lymphocyte count (A) and antibody level (B) of two individuals that underwent retroviral-based gene therapy treatments. A control (individual without X-linked SCID) is used as the reference of average lymphocyte count and antibody level in a healthy person.

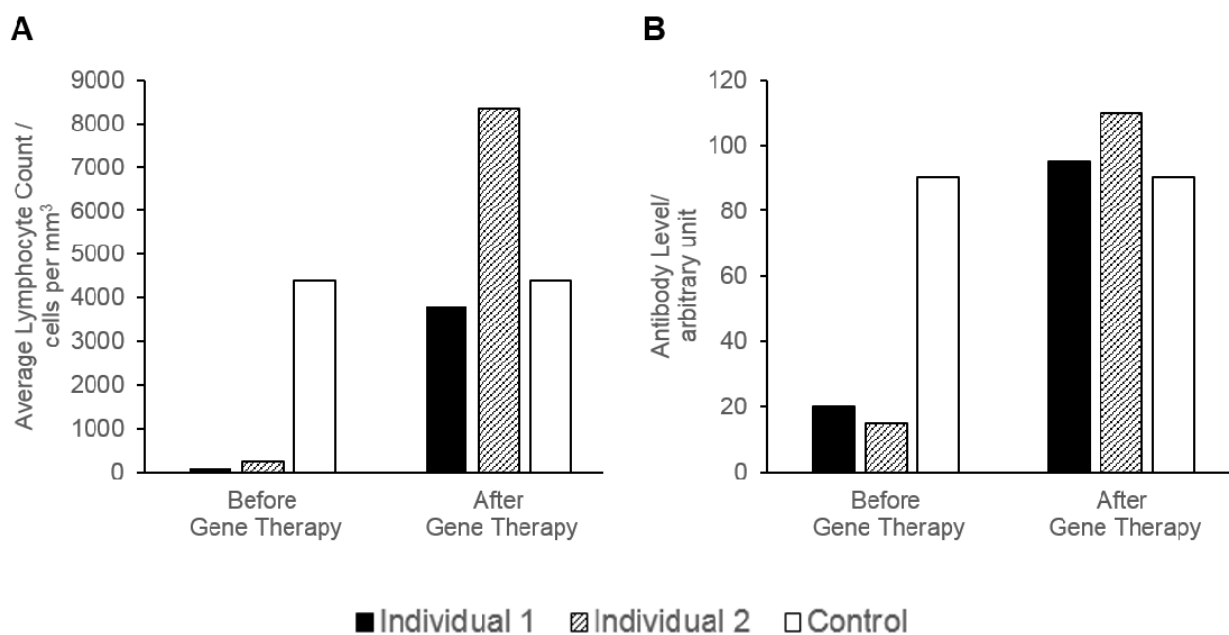


Fig. 3.2

(c) With reference to Fig. 3.2A and Fig. 3.2B,

(i) account for the effectiveness of gene therapy on Individual 1; [4]

(ii) suggest and explain the potential safety concern in Individual 2. [2]

(d) Discuss one ethical and one social consideration for the use of gene therapy. [2]

[Total: 14]

4 Planning Question

The yield of crop plants such as the onion, *Allium cepa*, is reduced by the presence of relatively high concentrations of salts, particularly sodium chloride, in the soil. There are, however, salt-tolerant varieties of onion, which can be grown successfully in such places. One possible explanation for this is that the fluid in the cell vacuoles of plants of the tolerant varieties differs in solute potential from that of non-tolerant varieties.

To measure the solute potential of the cell vacuole, one can make use of the concept of incipient plasmolysis. Incipient plasmolysis is the point where 50% of plant cells are plasmolysed.

Plan an investigation to find out whether the solute potentials of the cell vacuoles of epidermal cells of the bulbs of salt-tolerant onions differ from those of their non-tolerant counter parts.

You are provided with:

- 1.0 mol dm⁻³ sodium chloride solution
- two varieties of onion, one salt-tolerant and the other non-salt tolerant
- distilled water
- beakers of various sizes
- glass specimen tubes
- microscopic slides and coverslips
- measuring cylinders of various sizes
- dropping pipettes
- light microscope with low (x100), medium (x200) and high (x400) magnification and built in illumination
- stopwatch
- scalpel
- forceps
- glass rods for stirring
- mounted needles
- iodine in potassium iodide solution as a stain

Your plan should:

- have a clear and helpful structure such that the method you use is able to be repeated by anyone reading it,
- be illustrated by relevant diagram(s), if necessary,
- identify the independent and dependent variable,
- describe the method with the scientific reasoning used to decide the method so that the results are as accurate and reliable as possible,
- include layout of results tables and graphs with clear headings and labels,
- use the correct technical and scientific terms,
- include reference to safety measures to minimise any risks associated with the proposed experiment.

[Total: 12]

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[illegible]

Free-response question

Write your answers to this question on the separate answer paper provided.

Your answers:

- should be illustrated by large, clearly labelled diagrams, where appropriate.
- must be in continuous prose, where appropriate.
- must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

5 (a) Describe features of embryonic stem cells and adult stem cells that distinguish them from each other. [6]

(b) Outline the procedure for propagating plants by tissue culture technique. [6]

(c) Explain, with a named example, how genetic engineering improves the yield of crops. Include, in your explanation, two concerns associated with this genetically modified crop. [8]

[Total: 20]